

IP Addressing & Subnetting

Task – 1 : List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

Solution : IP Address Classes

IPv4 addresses were traditionally divided into **Class A, B, C, D, and E** based on their first octet. This is called **classful addressing**. Modern networks generally use **CIDR/subnetting** instead, but the classes are still important for understanding basic networking.

Class	First Octet Range	Address Range	Main Purpose	Real-World Example
A	1–126	1.0.0.0 – 126.255.255.255	Very large networks	Large organization/private network such as 10.0.0.0/8
B	128–191	128.0.0.0 – 191.255.255.255	Medium-sized networks	College or enterprise network such as 172.16.0.0/16
C	192–223	192.0.0.0 – 223.255.255.255	Smaller networks	Home/office LAN such as 192.168.1.0/24
D	224–239	224.0.0.0 – 239.255.255.255	Multicast	Video/audio multicast or routing protocols
E	240–255	240.0.0.0 – 255.255.255.255	Experimental/reserved	Experimental networking

Some addresses are reserved:

- **127.0.0.0/8** → Loopback
- **10.0.0.0/8** → Private Class A range
- **172.16.0.0/12** → Private Class B range
- **192.168.0.0/16** → Private Class C range

Task – 2 : Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Hint : Show your working steps for each calculation.

Solution : **192.168.10.45/24**

Given

- **IP Address:** 192.168.10.45
- **Subnet Mask:** 255.255.255.0
- **CIDR:** /24
- **Network bits:** 24
- **Host bits:** 8

Step 1: Network Address

IP: 192.168.10.45

Mask: 255.255.255.0

Since /24 means the first 3 octets are the network:

Network Address = 192.168.10.0

Step 2: Broadcast Address

All 8 host bits are set to 1:

192.168.10.11111111 = 192.168.10.255

Broadcast Address = 192.168.10.255

Step 3: Valid Host Range

Network: 192.168.10.0

First Host: 192.168.10.1

Last Host: 192.168.10.254

Broadcast: 192.168.10.255

Answer:

Network: 192.168.10.0

Broadcast: 192.168.10.255

Valid Hosts: 192.168.10.1 – 192.168.10.254

Task – 3 : Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Solution : **Zomato-Style Startup Subnetting**

The startup has the network : 10.0.0.0/24

It needs **4 separate subnets**:

1. Delivery Partners
2. Restaurants
3. Customers
4. Admins

Step 1: Determine Required Subnet Bits

We need at least **4 subnets**.

Formula : $2^n \geq 4$

Therefore : $2^2 = 4$

We need to borrow **2 bits** from the host portion.

Original : /24

New prefix : $24 + 2 = /26$

New Subnet Mask /26 = 255.255.255.192

Each subnet contains : $2^6 = 64$ addresses

Usable hosts : $64 - 2 = 62$ hosts

Subnetting Plan

Department	Network Address	Subnet Mask	Valid Host Range	Broadcast
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

Task – 4 : You are given the following scenario : A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Constraint : Show your calculation steps and final answers.

Solution : **College Wi-Fi Network**

- **Network** : 172.16.0.0/16
- **Required** : At least **20 subnets**

Step 1: Borrow Bits

Formula : $2^n \geq 20$

5 bits are needed.

Step 2: New Prefix & Mask

/16 + 5 = /21

New subnet mask: 255.255.248.0

Step 3: Number of Subnets

$2^5 = 32$ subnets

So, **32 subnets** can be created.

Step 4: Hosts Per Subnet

Host bits = $32 - 21 = 11$

$2^{11} = 2048$ addresses

$2048 - 2 = 2046$ usable hosts

Task – 5 : A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

Solution : **Invalid IP Address 192.168.1.300**

The IP address 192.168.1.300 is **invalid**.

Why?

An IPv4 address contains **four octets**, and every octet must have a value between 0 and 255

The address has : 192 . 168 . 1 . 300

The fourth octet is **300**, which is greater than the maximum allowed value of **255**.

What Should It Look Like?

If the device is using the common subnet : 192.168.1.0/24

then valid host addresses are normally:

192.168.1.1, 192.168.1.2, 192.168.1.3 ... 192.168.1.254

So an example of a valid address would be **192.168.1.100**

The addresses **192.168.1.0** and **192.168.1.255** are normally reserved for the network and broadcast addresses respectively in a /24 network.